MICROSOFT CERTIFIED: AZURE FUNDAMENTALS

AZ-900 Vibe Planner
Full Project Implementation Plan

An AI-powered adaptive study platform designed to accelerate candidate exam readiness through personalized schedules, active recall micro-drills, and dynamic workload re-balancing.

PROJECT NAME	**TARGET EXAMINATION**
AZ-900 Vibe Planner (CloudPrep AI)	Microsoft Azure Fundamentals (AZ-900)
ARCHITECTURE	**DOCUMENT VERSION**
Next.js 16 + React 19 + TypeScript + Supabase + GenAI	v1.2 — Production Release & Integration

Executive Brief: This document outlines the end-to-end technical architecture, exam syllabus mapping, implementation phases, backend API routes, UI/UX interaction workflows, and testing strategy for the AZ-900 Vibe Planner suite.

1. Executive Summary & Core Objectives

The **AZ-900 Vibe Planner** is a purpose-built exam preparation command center designed to eliminate the friction, study burnout, and schedule rigidity commonly experienced by cloud certification candidates. Instead of static checklists or unstructured video courses, the platform uses an intelligent 3-tier micro-learning model combined with a reactive *"Life Happens" Re-balancer*.

Key Success Metrics & Pillars

- **Syllabus Fidelity:** 100% mapped to Microsoft's official AZ-900 exam objectives and domain percentage weights.
- **Bite-Sized Efficiency:** Structured daily sessions (15–60 mins) broken into 3 actionable blocks: *Read & Watch* (15m), *Hands-On Sandbox* (10m), and *Active Recall* (5m).
- **Adaptive Workload Management:** One-click schedule re-balancing that redistributes missed study sessions across remaining days without moving the candidate's exam deadline.
- **Active Recall Retention:** Integrated knowledge check questions with candidate draft inputs, interactive trainer reveals, and core exam takeaways.
- **Zero-Barrier Instant Demonstration:** Instant judge demo mode pre-populating a realistic 7-day plan with completed metrics and live progress bars.

2. Official AZ-900 Exam Domain Weights

The planner weights all readiness calculations, mastery progress bars, and schedule distributions directly according to Microsoft's official exam scoring guidelines:

Domain Name	Official Weight	Key Exam Topics Covered	Platform Visual Badge
Cloud Concepts	**25–30%**	Shared Responsibility Model, IaaS vs PaaS vs SaaS, Public/Private/Hybrid Cloud, CapEx vs OpEx economics.	**Sky Blue Badge (25–30%)**
Azure Architecture & Services	**35–40%** ★ **Highest Weight**	Regions, Availability Zones, Region Pairs, Compute (VMs, Scale Sets, App Services, Functions, ACI/AKS), Networking (VNets, NSGs, ExpressRoute), Storage Tiers (Hot/Cool/Archive, LRS/ZRS/GRS), Databases.	**Amber Badge (35–40% High Priority)**
Azure Management & Governance	**30–35%**	Microsoft Entra ID, Zero Trust Architecture, MFA, RBAC permissions, Azure Policy, Resource Locks, Azure Advisor, Microsoft Defender for Cloud.	**Emerald Badge (30–35%)**
Final Review, SLAs & Pricing	**Composite & Readiness**	Azure Pricing Calculator, TCO Calculator, Cost Management & Budgets, Service Level Agreements (SLAs), Composite SLA math, Mock Exam Drills.	**Purple Badge (Readiness Drill)**

3. System Architecture & Tech Stack

The platform is built using a modern full-stack TypeScript architecture optimized for high performance, edge routing, offline fault-tolerance, and AI integration.

Tier	Technology	Role & Responsibilities
Frontend Framework	**Next.js 16 (App Router) + React 19**	Server components for SEO & static pages; interactive client components for real-time reactivity and local state.
Language	**TypeScript 5**	Strict type definitions across all study plan models, AI generators, API contracts, and user preferences.
Styling & Design System	**Tailwind CSS v4 + Glassmorphism**	Deep slate dark dashboard (#0B0F19), Azure Blue accent (#0078D4), backdrop-filter glass panels, custom responsive grids.
Micro-Interactions	**canvas-confetti + Lucide Icons**	Multi-burst celebration confetti upon completing study days; full iconography across Azure services.
State & Persistence	**Local Storage + Supabase Client**	Dual-layer persistence: immediate client-side offline storage backed by Supabase cloud storage synchronization.
AI Engine	**Google Gemini SDK (@google/genai) + OpenAI SDK**	Dynamic plan synthesis, domain-weighted schedule adaptation, and real-time AI Study Buddy tutoring.

Application Navigation Structure

- `/` — Executive Command Center (KPIs, days countdown, readiness gauge, feature quick links).
- `/planner` — Interactive Study Planner (Onboarding wizard, filter tabs, 7-day micro-learning cards, re-balancer).
- `/quizzes` — Timed interactive practice exam questions with domain category filters and score tracking.
- `/flashcards` — Active recall flip cards for high-yield exam facts (SLA math, redundancy tiers, cloud models).
- `/study-buddy` — AI Azure tutor providing tailored analogies and exam trap explanations.
- `/readiness` — Detailed domain analytics, radar mastery bars, and exam-day readiness checklist.
- `/settings` — Candidate exam target date, study capacity, and cloud synchronization controls.

4. Core Feature & Component Specifications

Feature 1: 30-Second Smart Onboarding Wizard

IMPLEMENTED

Component: `components/OnboardingWizard.tsx`

- **Target Exam Date Picker:** Real-time dynamic calculation of "*X Days Remaining*".
- **Study Capacity Selector:** Interactive pill buttons for 15, 30, 45, and 60+ minutes per day with a toggle for "*2x time on weekends*".
- **Domain Confidence Sliders:** 1-to-5 scale range sliders for Cloud Concepts, Architecture & Services (highlighted as high priority), and Management & Governance.
- **AI Synthesis Multi-Step Loader:** 4-step progressive animation demonstrating domain analysis and curriculum customization before revealing the roadmap.
- **Instant Judge Demo CTA:** One-click load of a pre-populated plan with Day 1 & Day 2 completed for instant showcase.

Feature 2: Executive KPI Header & "Life Happens" Re-balancer

IMPLEMENTED

Component: `components/ReadinessHeader.tsx`

- **Countdown Card:** Prominent numerical countdown to candidate exam date.
- **Weighted Readiness Score (%):** Real-time circular SVG gauge applying official domain weights (27.5% + 37.5% + 32.5% + 2.5%).
- **Domain Mastery Progress:** 3 distinct progress bars with domain badges and percentage indicators.
- **The "Life Happens" Re-balancer:** When triggered, detects missed sessions, smoothly redistributes remaining workload forward, updates card status to `rebalanced`, and presents an interactive toast.

Feature 3: Micro-Learning Daily Action Cards

IMPLEMENTED

Component: `components/DailyTaskCard.tsx`

- **Step 1: Read & Watch (15m):** Objective statement paired with direct "*Search on MS Learn ↗*" linking directly to official Microsoft documentation.
- **Step 2: Hands-On Sandbox (10m):** Step-by-step portal exercises (e.g. creating NSG rules, comparing VM sizes, configuring resource locks).
- **Step 3: Active Recall Checkpoint (5m):** Exam-style question, draft input box with quick save, and a reveal toggle that unveils trainer explanations and key exam takeaways.
- **Card Footer Toggle:** "*Mark Day Complete ✅*" with celebratory multi-burst confetti and automatic recalculation of overall readiness metrics.

5. Multi-Phase Implementation Roadmap

Phase	Milestone Focus	Deliverables & Scope	Status
Phase 1	Project Foundation & Design System	Next.js 16 setup, Tailwind CSS v4 design tokens, glassmorphism CSS utilities, Azure color palette, Lucide icon library.	COMPLETED
Phase 2	Core Study Planner & Onboarding	Interactive Onboarding Wizard, 7-day realistic mock curriculum, domain weight sliders, filter bar, daily action cards, confetti micro-interactions.	COMPLETED
Phase 3	Dynamic Re-balancer & Persistence	"Life Happens" redistribution algorithm, localStorage state sync, status transitions (completed, up-next, pending, rebalanced).	COMPLETED
Phase 4	Command Center & Multi-Page Suite	Dashboard overview page (`/`), flashcards module (`/flashcards`), practice quizzes (`/quizzes`), readiness analytics (`/readiness`), settings (`/settings`).	COMPLETED
Phase 5	AI Study Buddy & Backend API	API endpoints (`/api/generate-plan`, `/api/tasks/[id]`, `/api/rebalance`), Google Gemini & OpenAI integrations, AI Study Buddy chat (`/study-buddy`).	COMPLETED
Phase 6	Cloud DB Sync & Polish	Supabase task synchronization, offline-first fallback hardening, mobile drawer navigation, comprehensive testing.	COMPLETED

6. Quality Assurance & Verification Matrix

> **Verification Standard:** All production builds must pass with zero TypeScript errors, zero ESLint warnings, and 100% responsive layout across mobile (375px), tablet (768px), and desktop (1280px+).

Verification Area	Test Method	Result
TypeScript Build	`npm run build` executing Turbopack typechecking and route compilation across all 13 routes.	**Passed (0 Errors)**
Onboarding Flow	Browser subagent automated testing: date picker, capacity pills, domain confidence sliders, judge demo loader.	**Verified & Clickable**
Re-balancer Reaction	Simulated missed study day: verification of pending day detection, date offset, and toast notification appearance.	**Verified & Clickable**
Active Recall Toggle	Text input persistence, trainer explanation reveal, key takeaway callout.	**Verified & Clickable**
State Persistence	Browser refresh test verifying state preservation across page reloads via localStorage.	**Verified**